

FINAL PROJECT REPORT

CREDIT CARD FRAUD DETECTION

USING AIML

Project Domain: Artificial Intelligence & Machine Learning

Tools Used: Python, Google Colab, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Imbalanced-learn, XGBoost, SHAP, Joblib

Abstract

Credit card fraud is one of the major challenges faced by financial institutions worldwide due to the rapid increase in digital transactions. Fraudulent transactions cause financial losses and reduce customer trust. This project develops a machine learning-based fraud detection system capable of identifying fraudulent credit card transactions using historical transaction data. Multiple machine learning algorithms were trained and evaluated after performing data preprocessing, exploratory data analysis, feature engineering, and handling class imbalance using SMOTE. The best-performing model was selected based on comprehensive evaluation metrics and saved for future prediction. The project demonstrates how machine learning can improve fraud detection while supporting future deployment into banking and financial applications.

1. Introduction

With the increasing adoption of online payments and digital banking, credit card fraud has become a serious concern. Traditional rule-based fraud detection systems are limited because fraudsters continuously develop new techniques to bypass predefined rules.

Machine learning provides an intelligent solution by learning transaction patterns from historical data and automatically identifying suspicious

transactions. This project builds a fraud detection model capable of classifying transactions as either fraudulent or genuine while minimizing false alarms.

2. Problem Statement

The objective of this project is to design a machine learning model that accurately detects fraudulent credit card transactions despite the highly imbalanced nature of the dataset. The solution should improve fraud detection accuracy while reducing false positives and false negatives.

3. Objectives

- Develop a machine learning model for fraud detection.
 - Perform data cleaning and preprocessing.
 - Analyze transaction patterns through exploratory data analysis.
 - Handle class imbalance using SMOTE.
 - Compare multiple classification algorithms.
 - Evaluate model performance using standard metrics.
 - Save the trained model for future predictions.
 - Build a prediction system for new transactions.
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4. Dataset Description

Dataset Name: Credit Card Fraud Detection

Source: Kaggle

Dataset Information

- Total Records: 284,807
 - Total Features: 30 input features
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- Target Column: Class

Target Labels

- 0 → Genuine Transaction
- 1 → Fraudulent Transaction

Feature Description

- Time – Time elapsed between transactions.
- Amount – Transaction amount.
- V1-V28 – Anonymized features generated using Principal Component Analysis (PCA).
- Class – Target variable.

The original transaction attributes were transformed into PCA components to preserve customer privacy.

5. Methodology

The project followed the following workflow:

1. Environment Setup
2. Data Loading
3. Data Cleaning
4. Exploratory Data Analysis
5. Feature Engineering
6. Feature Scaling
7. Handling Imbalanced Data using SMOTE
8. Train-Test Split
9. Model Training

10. Model Evaluation
 11. Model Comparison
 12. Explainable AI (SHAP)
 13. Model Saving
 14. Prediction System
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6. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset.
- Verified dataset dimensions.
- Checked column names and data types.
- Identified missing values.
- Removed duplicate records.
- Prepared a clean dataset.
- Separated features and target variable.
- Applied StandardScaler and RobustScaler.
- Balanced the training data using SMOTE.

Observation

The dataset contained no missing values. Duplicate records were removed, and SMOTE effectively balanced the minority fraud class, enabling better model learning.

7. Exploratory Data Analysis

Several visualizations were generated to understand transaction behavior.

Fraud Distribution

The dataset contains significantly more genuine transactions than fraudulent ones, confirming a severe class imbalance.

Transaction Amount Distribution

Most transactions involve relatively small amounts, with a small number of high-value outliers.

Transaction Time Distribution

Transactions are spread throughout the recorded period, indicating continuous transaction activity.

Correlation Heatmap

The PCA-transformed features exhibit varying degrees of correlation. Features such as V14, V17, and V12 contribute significantly to fraud detection.

8. Feature Engineering

The dataset was prepared for machine learning by:

- Separating input features (X)
 - Separating target labels (y)
 - Scaling numerical data
 - Applying SMOTE to balance the classes
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9. Machine Learning Models

The following classification algorithms were implemented:

Logistic Regression

Used as the baseline linear classification model.

Decision Tree

Provides nonlinear decision boundaries and interpretable decision rules.

Random Forest

An ensemble learning algorithm combining multiple decision trees for improved accuracy and robustness.

XGBoost

A gradient boosting algorithm that sequentially improves prediction performance.

10. Model Evaluation

Each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score

Additional evaluation techniques included:

- Confusion Matrix
- ROC Curve
- Precision-Recall Curve

These metrics provide a comprehensive understanding of model performance, especially for imbalanced datasets.

11. Model Comparison

All trained models were compared based on their evaluation metrics.

Findings

- Logistic Regression provided a strong baseline.
- Decision Tree learned complex patterns but was more prone to overfitting.
- Random Forest demonstrated a strong balance between Precision, Recall, and ROC-AUC.
- XGBoost produced competitive results with higher computational cost.

The best-performing model was selected based on overall performance rather than accuracy alone.

12. Explainable AI (SHAP)

SHAP was used to explain the predictions of the selected model.

The SHAP analysis identified the most influential features contributing to fraud detection, improving model transparency and interpretability.

13. Model Saving

The final trained model and preprocessing scaler were saved using Joblib as:

- fraud_model.pkl
- scaler.pkl

These files allow future predictions without retraining the model.

14. Prediction System

A reusable prediction function was developed that:

- Accepts new transaction data.
 - Applies feature scaling.
 - Loads the saved model.
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- Predicts whether the transaction is Fraud or Genuine.
 - Returns the prediction along with the confidence score.
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15. Results

The project successfully developed a complete machine learning pipeline for fraud detection.

Major achievements include:

- Successful preprocessing of transaction data.
 - Effective handling of class imbalance using SMOTE.
 - Comprehensive exploratory data analysis.
 - Training and evaluation of multiple machine learning models.
 - Selection of the best-performing model.
 - Model persistence using Joblib.
 - Development of a prediction system for future deployment.
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16. Business Impact

The developed solution can help financial institutions by:

- Detecting fraudulent transactions more accurately.
 - Reducing financial losses.
 - Improving customer trust.
 - Supporting fraud investigation teams.
 - Enabling intelligent transaction monitoring.
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17. Limitations

- The dataset is highly imbalanced.
 - PCA-transformed features reduce interpretability.
 - The model was trained on historical data and may require periodic retraining.
 - Fraud patterns continuously evolve over time.
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18. Future Scope

Future improvements include:

- Hyperparameter optimization.
 - Deep learning-based fraud detection.
 - Real-time fraud monitoring.
 - Streamlit or Flask web application deployment.
 - Cloud deployment using AWS or Azure.
 - Continuous model retraining.
 - Advanced Explainable AI techniques.
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19. Final Findings

- The dataset is highly imbalanced, making fraud detection a challenging problem.
- SMOTE successfully balanced the training dataset and improved the model's ability to detect fraud.
- Multiple machine learning algorithms were trained and compared.
- The selected model demonstrated strong performance based on Precision, Recall, F1 Score, and ROC-AUC.

- The prediction system can classify new transactions as fraudulent or genuine after preprocessing.
 - The complete workflow demonstrates the practical application of machine learning in financial fraud detection.
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20. Conclusion

This project successfully developed a machine learning-based credit card fraud detection system using data preprocessing, exploratory data analysis, feature engineering, class balancing, and multiple classification algorithms. The models were evaluated using industry-standard metrics, and the best-performing model was selected for deployment. The trained model and preprocessing scaler were saved for future use, while the prediction function enables real-time transaction classification.

Overall, the project demonstrates how machine learning can significantly improve fraud detection, reduce financial risk, and support decision-making in modern banking systems. It also provides a scalable foundation for future deployment as a web application or real-time fraud monitoring solution.